

ChewML

Food Classification via AirPods IMU Sensors

Jonah Karstens · 4004353

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Motivation



63%

of eating detection
research uses accelerometers

(Scoping Review, npj Digital Medicine 2020)



~50 Hz

IMU data from commercial
earbuds (AirPods, Galaxy Buds)

(IMChew, ACM BodySys 2024)



0%

manual input required —
fully passive monitoring

(Project goal)

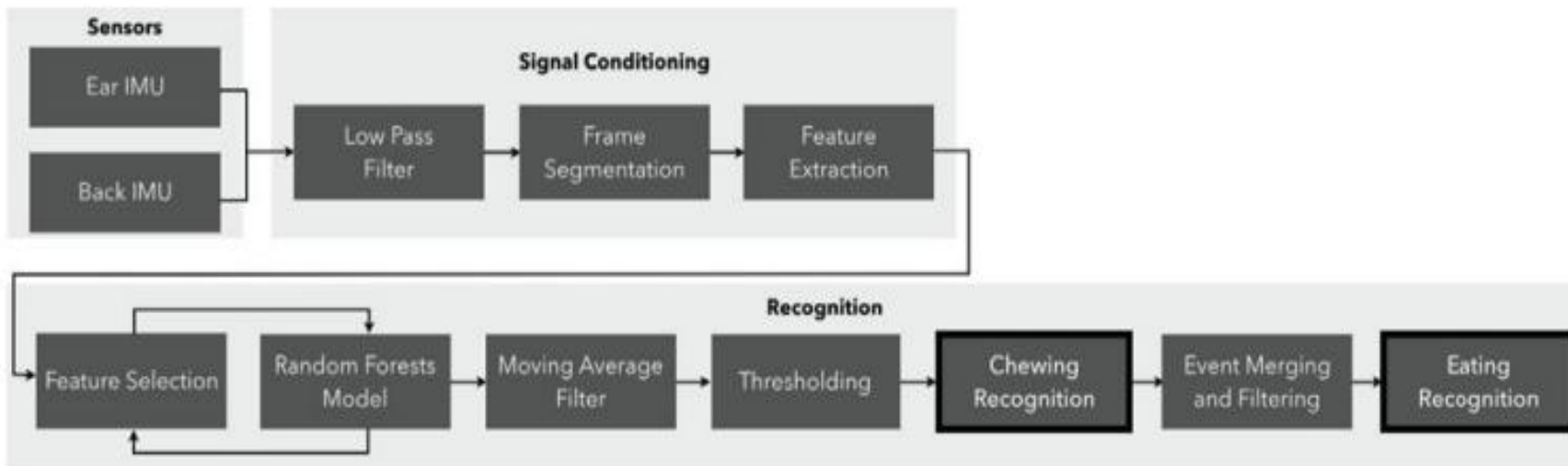
The temporalis muscle (linked to the jaw) sits right behind the ear — making in-ear IMU sensors ideally positioned to capture chewing motion.

Related Work

EarBit (UbiComp 2018)

PMC · [ncbi.nlm.nih.gov](https://pubmed.ncbi.nlm.nih.gov/)

Wearable in-ear sensors to detect eating episodes in unconstrained environments. Demonstrates that passively worn ear sensors can reliably detect eating vs. non-eating.



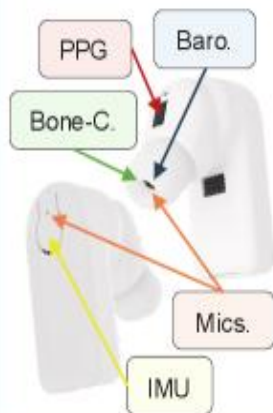
Related Work

CHOMP (arXiv 2026)

Multimodal earphone study

IMU + microphone + PPG across 11 foods and 20 participants. 97.7% leave-one-food-out accuracy. Detecting unhealthy chewing patters.

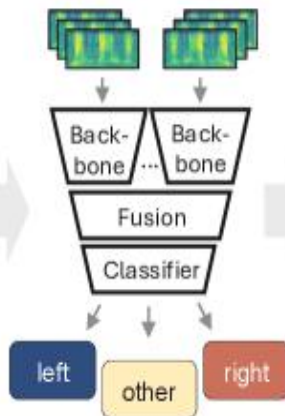
A. OpenEarable 2.0



B. Monitor Chewing



C. Detect Chewing Side



D. Chewing Side Analysis



Research Question

Can chewing patterns captured via in-ear IMU sensors (AirPods Pro) be used to classify different foods — and distinguish eating from not eating?

 Apfel

 Kaugummi

 Skyr

 Still

Data Collection



Dataset Overview

Class	Sessions	Duration	Sample Rate
Apfel	15	~60 s each	~50 Hz
Kaugummi	15	~60 s each	~50 Hz
Skyr	15	~60 s each	~50 Hz
Still	15	~60 s each	~50 Hz
Total	60	self-recorded	AirPods Pro

Sensor Channels

- Accelerometer X / Y / Z
- Gravity X / Y / Z
- Gyroscope X / Y / Z
- Pitch · Roll · Yaw

From Raw Signal to Features

Signal Preparation Example - Magnitude

1 Trim edges

First & last 5 seconds are cut — the recording starts and stops while already eating, so those edges are unreliable.

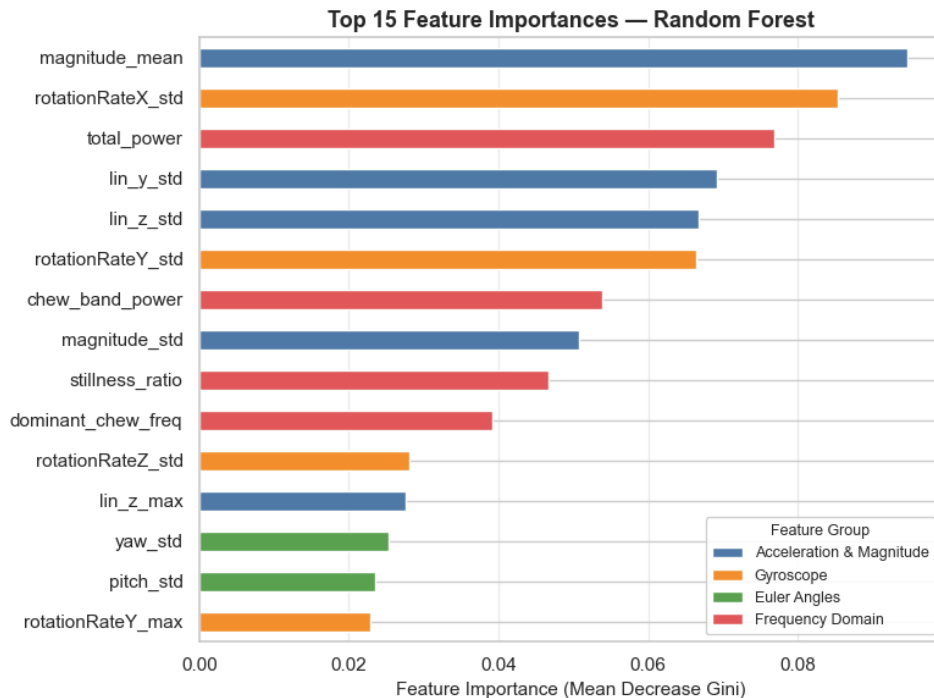
2 Remove gravity

The sensors always measure gravity (constant pull). We subtract it so only actual jaw movement remains.

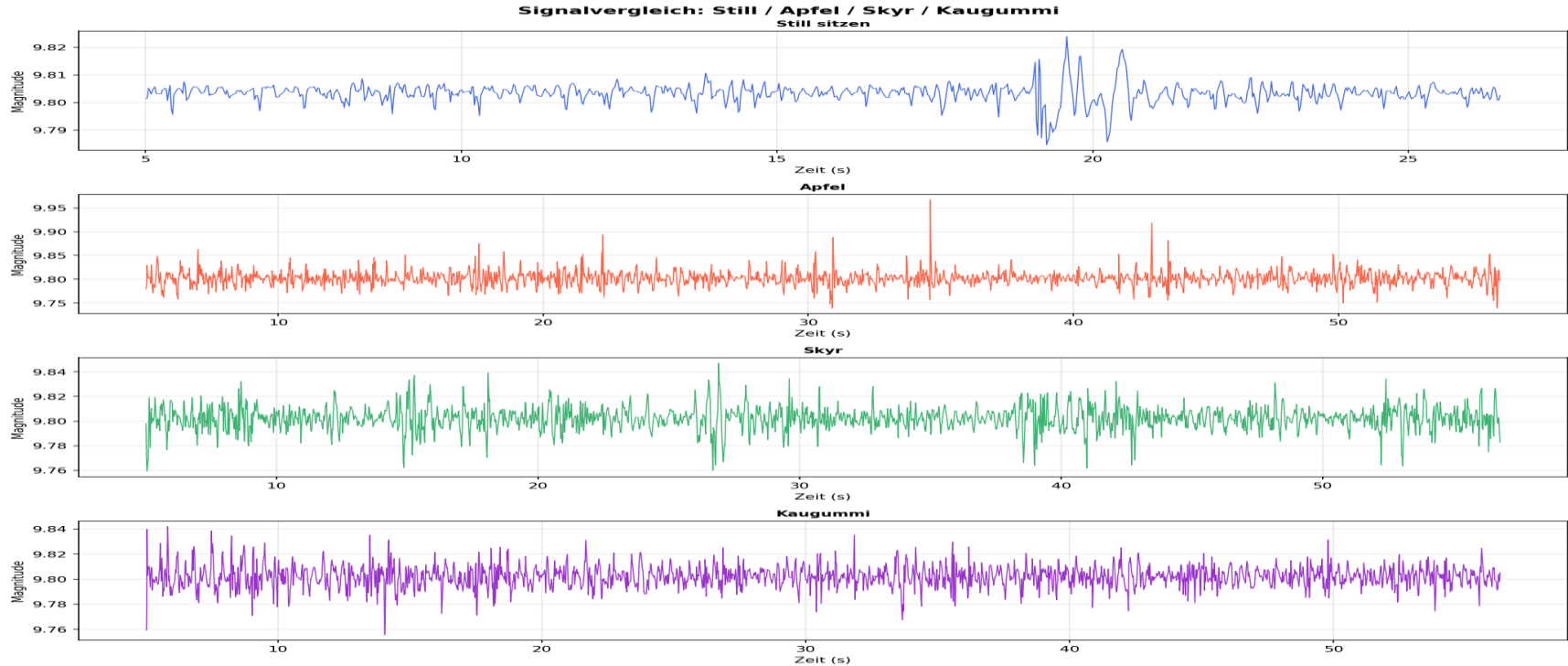
3 One signal per moment

X, Y, Z axes are combined into a single number: the total movement strength — independent of how the AirPods sits in the ear.

What is measured (37 values per recording)



First Data: Signal Comparison



Still — flat, near-zero variance **Apfel** — irregular chewing bursts **Skyr** — steady rhythmic motion **Kaugummi** — rapid low-amplitude oscillations

Approach 1: Low-Pass Filter

How it works

A 4th-order Butterworth filter (zero-phase) attenuates all frequencies above a cutoff. Tested cutoffs: 1, 2, 4, 6 Hz. The chewing band (0.5–6 Hz) lies below all tested cutoffs — the chewing signal remains intact.

Why it doesn't solve the problem

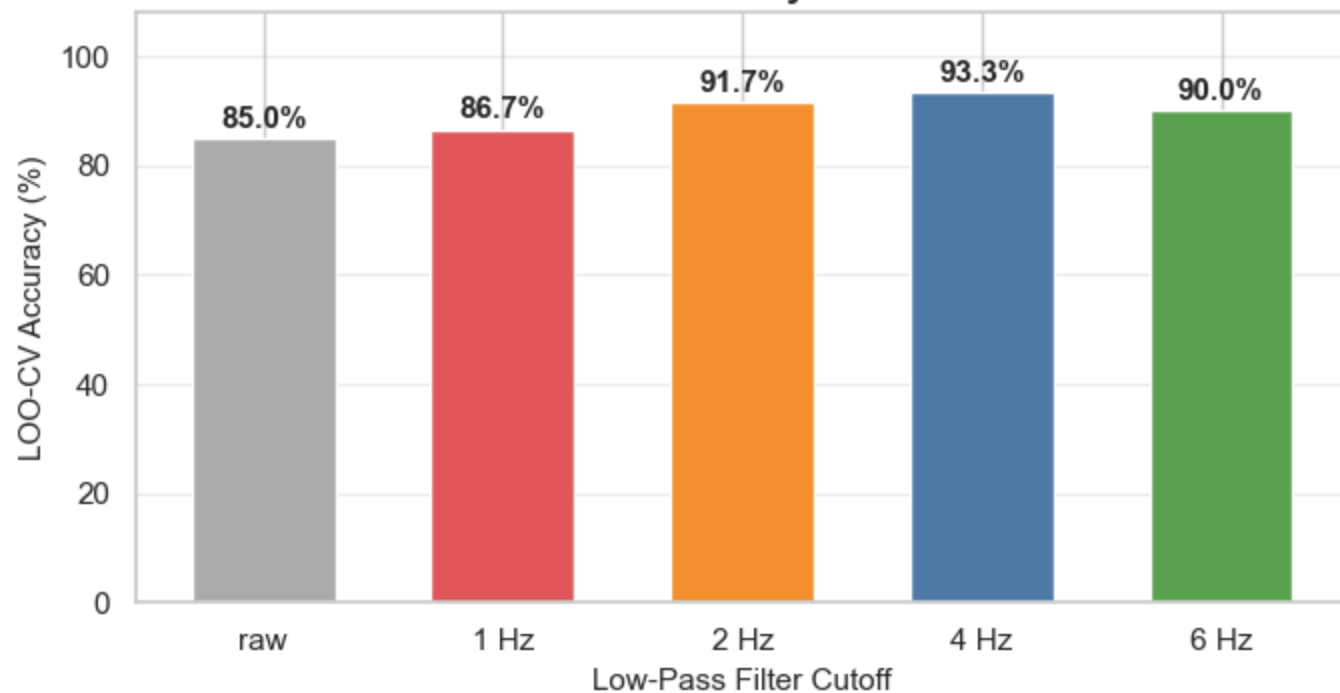
Head movements generate sustained signal bursts in the same frequency range as chewing $\sim$ (0.5–6 Hz). A low-pass filter cannot separate these overlapping frequency components.



Low-Pass Filter Comparison — Grid View (rows = filter cutoff, columns = class)



Classification Accuracy vs. Filter Cutoff



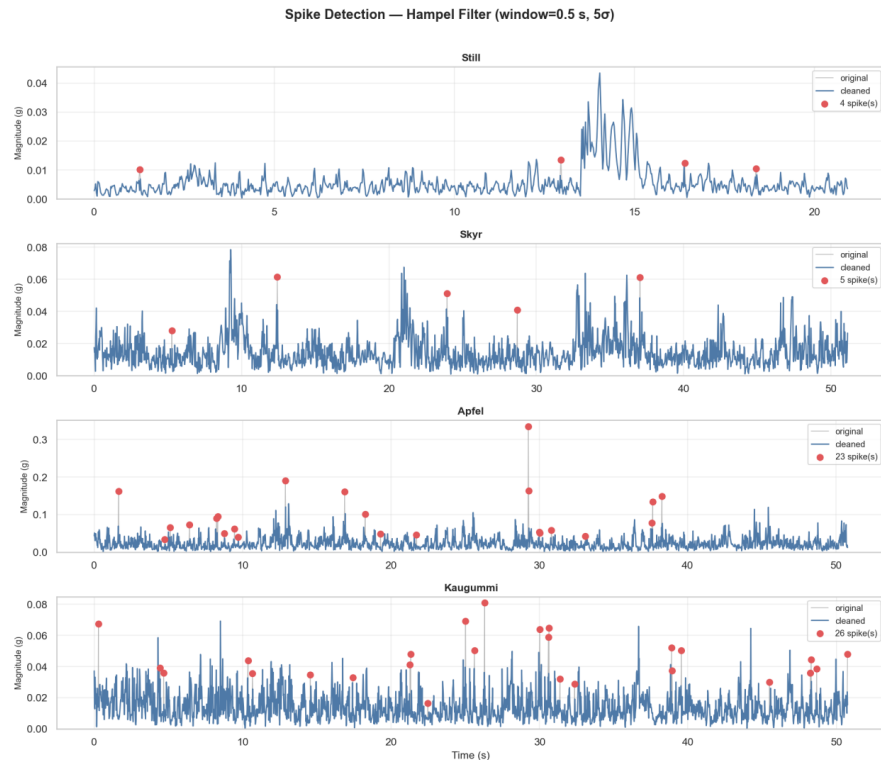
Approach 2: Hampel Filter

How it works

Detects isolated outliers via a rolling median: each sample is compared to its local median. Deviations $> 5 \times 1.4826 \times \text{MAD}$ (approx. 5 sigma) are flagged as spikes and interpolated. Window size: 0.5 s (25 samples).

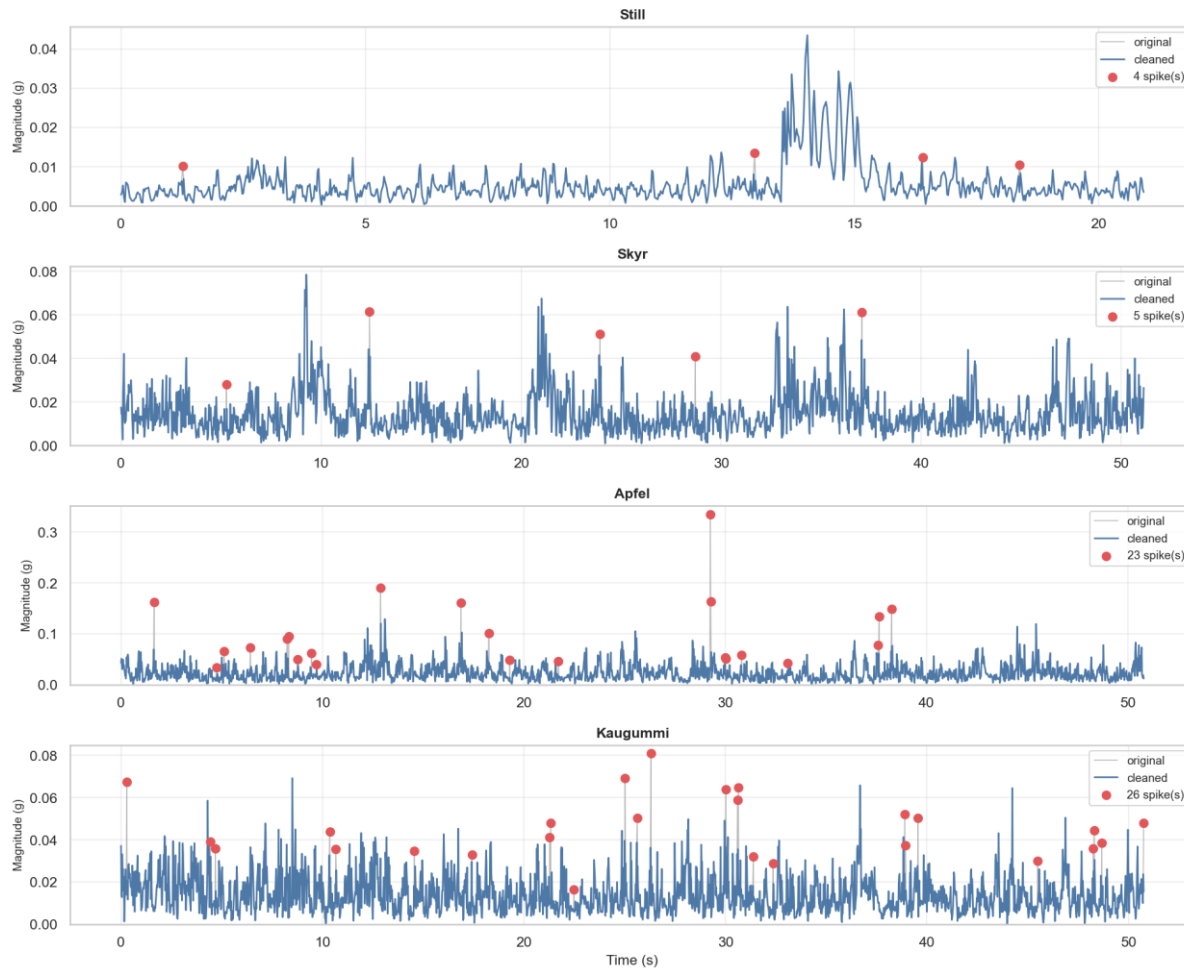
Why it doesn't solve the problem

Head movements are not isolated spikes — they last several seconds. The Hampel filter correctly removes only true single-sample outliers (5 or fewer per session) but leaves sustained movement phases completely untouched.

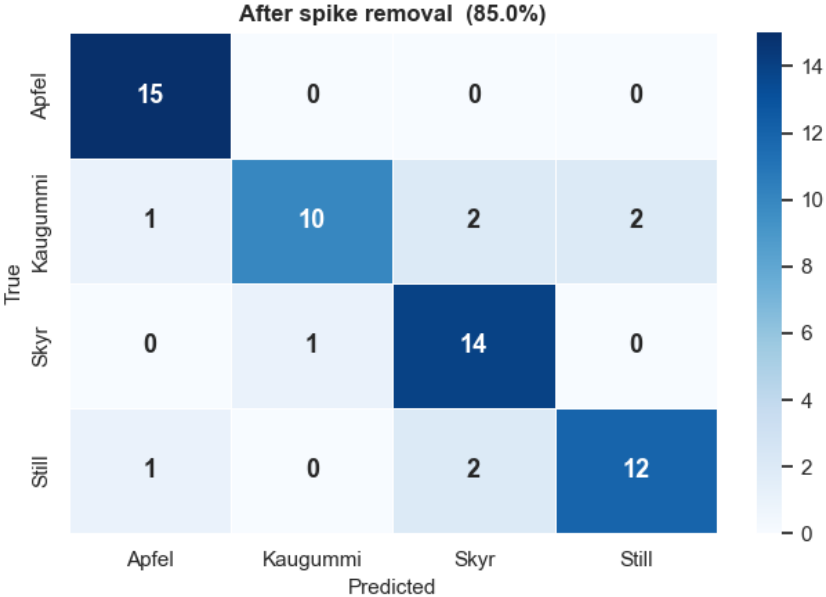
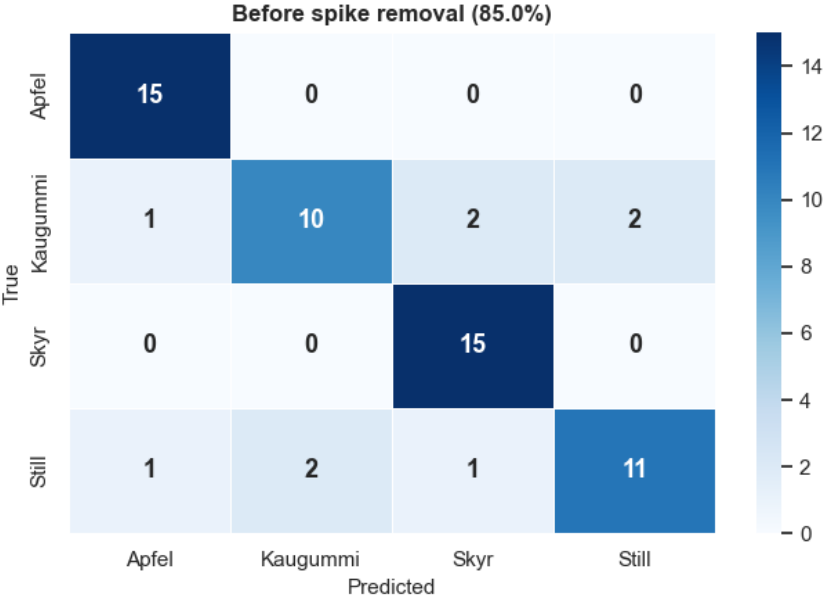


[Click to enlarge](#)

Spike Detection — Hampel Filter (window=0.5 s, 5σ)



Confusion Matrix — LOO-CV



Approach 3: Movement-Segment Exclusion

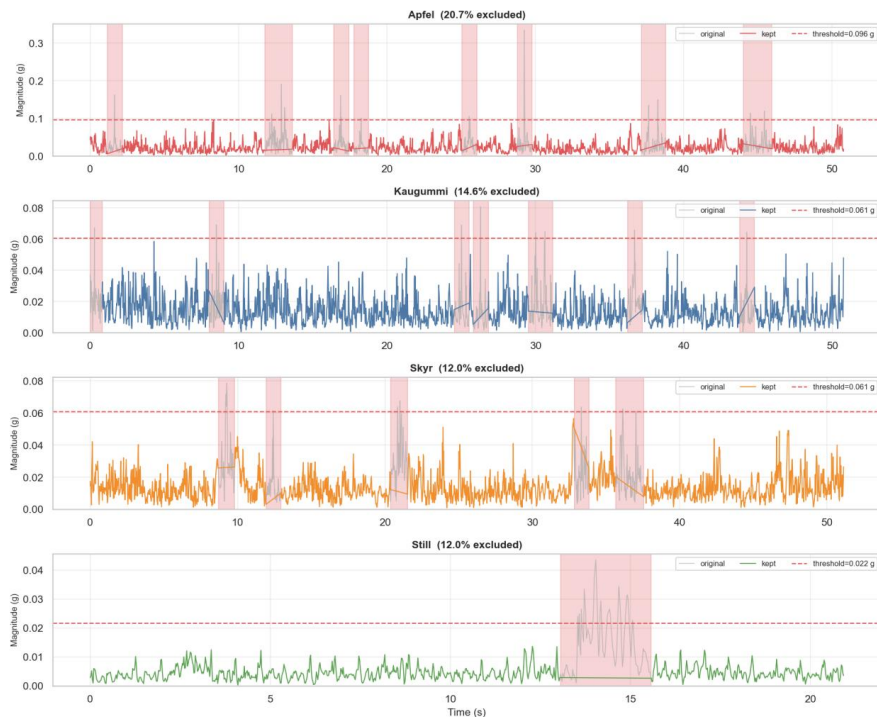
The Problem

Natural head movements while eating (looking around, nodding) create sustained high-amplitude signal bursts — indistinguishable from eating by simple time-domain statistics.

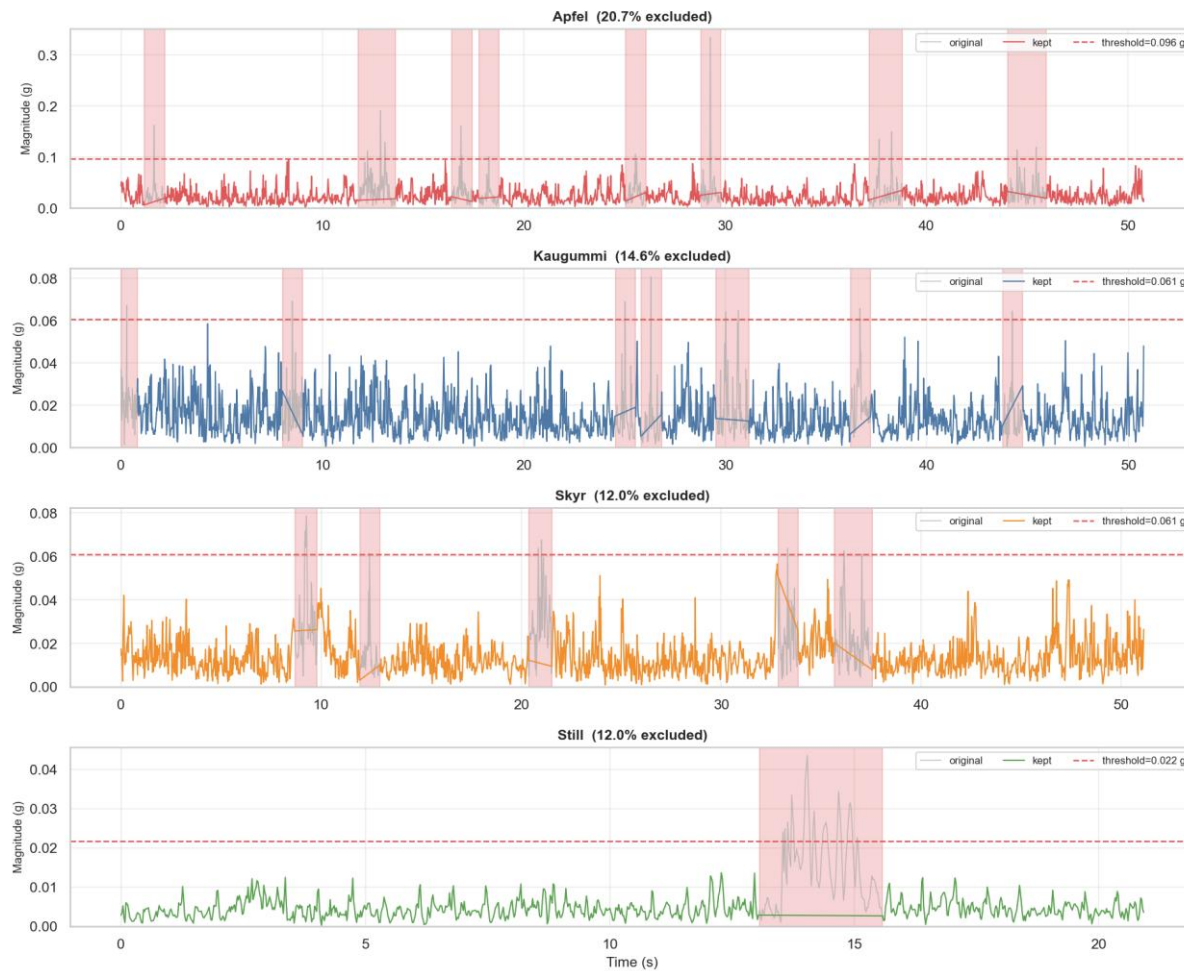
The Solution

Adaptive movement-segment exclusion: per-session threshold (5 x session median magnitude) removes head-movement bursts before feature extraction — without requiring class labels

Movement-Segment Exclusion — Adaptive Threshold (5 × session median)

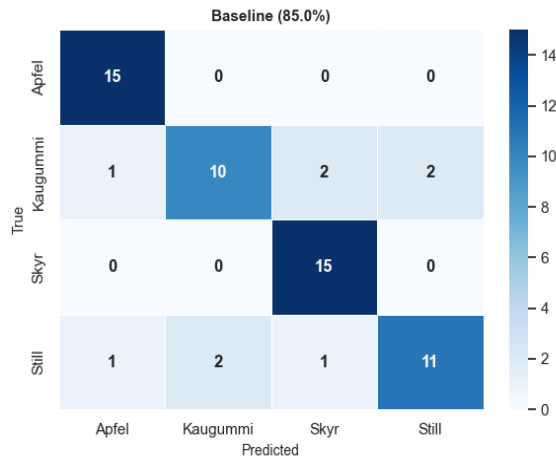


Movement-Segment Exclusion — Adaptive Threshold (5 × session median)



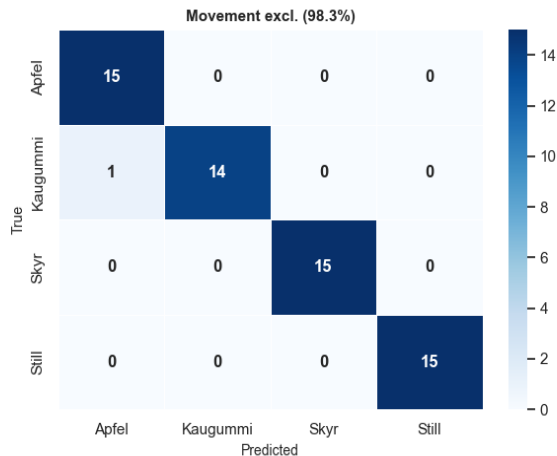
Results — LOO-CV Accuracy

Confusion Matrix Comparison — LOO-CV



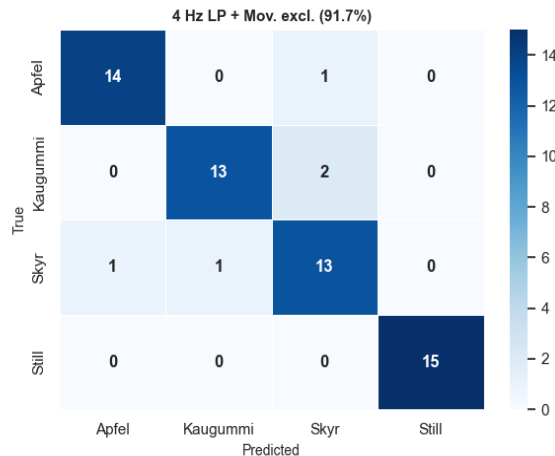
85%

Baseline
(no filter)



98,3%

Movement
exclusion



91.7%

4 Hz LP + Movement exclusion

Live Data Streaming — Proof of Concept

Already achieved: real-time sensor data transfer from AirPods via Sensor Logger directly to a local Python server over HTTP.



HTTP streaming: done



Live classifier: planned



Web dashboard: planned

Final vision: open app → eat → see real-time classification on screen. No manual input, no post-processing.

The Sensor Logger app supports HTTP endpoint streaming natively — no custom iOS code required.

Challenges & Limitations



Small Dataset

Currently ~60 sessions, mostly from one person. LOO-CV is valid but gives high-variance estimates. More data and more subjects needed for generalizable results.



Head Movement Overlap

Walking (~2 Hz) overlaps with the chewing frequency band (0.5–4 Hz). Simple movement exclusion works for sitting — more robust solution needed for real-world use.



Single-Subject Bias

All recordings so far from one person. Chewing style varies between individuals. Subject-independent evaluation is the next critical step.



Feature Selection Open

37 features extracted but not yet optimized. Which features survive subject-independence testing? This is the core open research question.

Plan Ahead

Week 7	Week 8	Week 9-10	Week 11-12
Dataset Expansion	Feature Selection & Model	Live Application	Final Evaluation
30 sessions / class (own recordings) - Multiple subjects (friends, family) - Evaluate subject-independence	- Motion-invariant feature design - Compare RF, SVM, kNN, MLP - Proper train/test split evaluation	- FastAPI streaming server - Sliding-window classifier - Web dashboard with real-time label	- Cross-subject testing - Final report & presentation - GitHub documentation

Target: subject-independent food classifier with real-time web application by end of semester.

Thank you.

Questions & Discussion

github.com/karstensjonah-design/ML4CS-group-ChewML

References

Bell, B. M., Alam, R., Alshurafa, N., Thomaz, E., Mondol, A. S., de la Haye, K., Stankovic, J. A., Lach, J., & Spruijt-Metz, D. (2020). Automatic, wearable-based, in-field eating detection approaches for public health research: a scoping review. *npj Digital Medicine*, 3(1), Article 38.

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